EduCAP™ Worksheet

Alabama Comprehensive Assessment Program

Bush Hills STEAM Academy
Full STEAM Ahead

ACAP Worksheet — Math G8

Subject: Math | Grade: 8 | DOK Level: 3 | Standard: 8.AF.5

Name: _____ Date: _____ Period: _____

[Visual Element for Question 1]

A table titled 'Celestial Body Data'. It has three columns: 'Celestial Body', 'Approximate Distance from Earth (km)', and 'Diameter Relative to Earth (Earth's Diameter = 1)'.
Row 1 (Headers): Celestial Body | Approximate Distance from Earth (km) | Diameter Relative to Earth (Earth's Diameter = 1)
Row 2: Moon | 3.84 x 10^5 | 2.73 x 10^-1
Row 3: Mars | 5.46 x 10^7 | 5.32 x 10^-1
Row 4: Jupiter | 5.88 x 10^8 | 1.12 x 10^1
Row 5: Pluto | 4.28 x 10^9 | 1.87 x 10^-1

1. An astronomer compiled data on the approximate distances of four celestial bodies from Earth and their diameters relative to Earth's diameter. Analyze the data in the table below and determine which statement accurately reflects a comparison of these celestial bodies.

A) The Moon is approximately 1.5 times larger in diameter than Pluto.
B) Jupiter's diameter is over 20 times greater than Mars's diameter.
C) Mars is closer to Earth than the Moon by a factor of approximately 1.4 x 10^2.
D) The distance from Earth to Pluto is more than 10 times the distance from Earth to Jupiter.

[Visual Element for Question 2]

A bar graph titled 'Bacterial Population Densities (cells per mm^3)'. The x-axis lists 'Bacteria Type' (A, B, C, D). The y-axis is 'Population Density' in scientific notation.
Bar for Bacteria A: 7.5 x 10^4
Bar for Bacteria B: 2.1 x 10^5
Bar for Bacteria C: 1.5 x 10^4
Bar for Bacteria D: 1.8 x 10^5

2. A microbiologist studied the average population densities of four different types of bacteria in a soil sample. The bar graph below illustrates these densities. Based on the data, select ALL statements that are true regarding the bacterial populations.

(Select ALL that apply)

%¡ A) Bacteria B has the highest population density among the four types.
%¡ B) The population density of Bacteria A is roughly 3.75 times greater than that of Bacteria C.
%¡ C) If Bacteria C's population density doubled, it would still be less than Bacteria A's population density.
%¡ D) The difference in population density between Bacteria B and Bacteria D is 6.0 x 10^4 cells per mm^3.
%¡ E) Bacteria D's population density is closer to Bacteria B's than to Bacteria A's.

[Visual Element for Question 3]

None

3. A scientist claims that a certain microscopic particle, with a mass of 9.11 x 10^-31 kg, is exactly 10^5 times lighter than another subatomic particle, which has a mass of 1.67 x 10^-27 kg. Evaluate the scientist's claim. Show all work and explain whether the claim is accurate or inaccurate, justifying your conclusion.

[Visual Element for Question 4]

None

4. The average distance from Earth to the Sun is approximately 1.5 x 10^8 kilometers. The distance light travels in one year (a light-year) is approximately 9.46 x 10^12 kilometers. A student claims that a light-year is over 60,000 times the average distance from Earth to the Sun. Which of the following evaluations best describes the student's claim?

A) The claim is accurate because (9.46 x 10^12) / (1.5 x 10^8) is approximately 6.3 x 10^4, which is 63,000.
B) The claim is inaccurate because (9.46 x 10^12) / (1.5 x 10^8) is approximately 6.3 x 10^3, which is 6,300.
C) The claim is accurate because 9.46 is much larger than 1.5, indicating a large difference.
D) The claim is inaccurate because 1.5 x 10^8 is a very large number, so 60,000 times that would be too immense for a light-year.

[Visual Element for Question 5]

A simplified cross-section diagram of a plant cell. The outer boundary of the cell is represented by a large oval.
Inside, a smaller, bean-shaped structure representing a mitochondrion is drawn.
An arrow labeled 'Plant Cell Diameter: 5.0 x 10^-5 m' spans the width of the large oval.
An arrow labeled 'Mitochondrion Diameter: 8.0 x 10^-7 m' spans the width of the bean-shaped structure.

5. Consider the diagram below, which illustrates a cross-section of a plant cell and one of its organelles, a mitochondrion. The approximate diameter of the plant cell is 5.0 x 10^-5 meters, and the approximate diameter of a mitochondrion within it is 8.0 x 10^-7 meters. Explain the relationship between the size of the plant cell and the mitochondrion. Specifically, determine how many times larger the plant cell is than the mitochondrion and justify the significance of this size difference for cellular function.

Answer Key

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1. B — To evaluate the statements, we compare the given values:
A: Moon diameter (2.73 x 10^-1 = 0.273) / Pluto diameter (1.87 x 10^-1 = 0.187) "H 1.46. So, A is incorrect.
B: Jupiter diameter (1.12 x 10^1 = 11.2) / Mars diameter (5.32 x 10^-1 = 0.532) "H 21.05. Since 21.05 is over 20, B is correct.
C: Mars distance (5.46 x 10^7) is much greater than Moon distance (3.84 x 10^5), so Mars is further, not closer. So, C is incorrect.
D: Pluto distance (4.28 x 10^9) / Jupiter distance (5.88 x 10^8) = (4.28 / 5.88) x 10^(9-8) "H 0.728 x 10 = 7.28. Since 7.28 is not more than 10, D is incorrect.

2. A,C,E — Let's convert all values to a common exponent or standard form for easier comparison:
Bacteria A: 7.5 x 10^4 = 75,000
Bacteria B: 2.1 x 10^5 = 210,000
Bacteria C: 1.5 x 10^4 = 15,000
Bacteria D: 1.8 x 10^5 = 180,000

A: Bacteria B (210,000) is indeed the highest population density. (True)
B: Ratio of A to C = (7.5 x 10^4) / (1.5 x 10^4) = 7.5 / 1.5 = 5. The statement says 'roughly 3.75 times'. (False)
C: Doubled C = 2 * (1.5 x 10^4) = 3.0 x 10^4 = 30,000. Is 30,000 less than A (75,000)? Yes. (True)
D: Difference between B and D = (2.1 x 10^5) - (1.8 x 10^5) = (2.1 - 1.8) x 10^5 = 0.3 x 10^5 = 3.0 x 10^4 = 30,000. The statement says 6.0 x 10^4. (False)
E: Distance D to B = |1.8 x 10^5 - 2.1 x 10^5| = 0.3 x 10^5.
Distance D to A = |1.8 x 10^5 - 0.75 x 10^5| = 1.05 x 10^5.
Since 0.3 x 10^5 < 1.05 x 10^5, D is closer to B. (True)

3. [Short Response]

Sample Response:
To evaluate the scientist's claim, we need to compare the mass of the subatomic particle (1.67 x 10^-27 kg) to the mass of the microscopic particle (9.11 x 10^-31 kg).
If the microscopic particle is 10^5 times lighter, then the subatomic particle's mass should be 10^5 times greater than the microscopic particle's mass.
Let's calculate 10^5 times the mass of the microscopic particle:
(9.11 x 10^-31 kg) * 10^5 = 9.11 x 10^(-31 + 5) kg = 9.11 x 10^-26 kg.
Now, we compare this calculated value (9.11 x 10^-26 kg) to the given mass of the subatomic particle (1.67 x 10^-27 kg).
To compare them easily, let's put them in the same power of 10:
9.11 x 10^-26 kg = 91.1 x 10^-27 kg.
Comparing 91.1 x 10^-27 kg with 1.67 x 10^-27 kg, we see that 91.1 is significantly larger than 1.67.
Alternatively, we can find the ratio of the subatomic particle's mass to the microscopic particle's mass:
Ratio = (1.67 x 10^-27 kg) / (9.11 x 10^-31 kg)
Ratio = (1.67 / 9.11) x 10^(-27 - (-31))
Ratio "H 0.1833 x 10^4
Ratio "H 1833.
The ratio is approximately 1833, not 10^5 (which is 100,000).
Therefore, the scientist's claim is inaccurate. The subatomic particle is approximately 1833 times heavier, not 100,000 times heavier (or 10^5 times heavier).

4. A — To evaluate the student's claim, we need to find the ratio of a light-year to the Earth-Sun distance:
Ratio = (9.46 x 10^12 km) / (1.5 x 10^8 km)
Ratio = (9.46 / 1.5) x 10^(12 - 8)
Ratio = 6.306... x 10^4
Ratio = 63,060.
Since 63,060 is indeed 'over 60,000', the student's claim is accurate. Option A correctly performs this calculation and provides the correct conclusion. Option B has an incorrect exponent for the result. Options C and D rely on vague estimations rather than precise calculation.

5. [Short Response]

Sample Response:

To determine how many times larger the plant cell is than the mitochondrion, we need to divide the diameter of the plant cell by the diameter of the mitochondrion.

Plant Cell Diameter = 5.0 x 10^-5 m

Mitochondrion Diameter = 8.0 x 10^-7 m

Ratio = (5.0 x 10^-5) / (8.0 x 10^-7)

Ratio = (5.0 / 8.0) x 10^(-5 - (-7))

Ratio = 0.625 x 10^2

Ratio = 62.5

The plant cell is 62.5 times larger in diameter than the mitochondrion.

The significance of this size difference for cellular function is that the cell is a complex system that must contain many different organelles, each with specialized functions. Mitochondria are responsible for cellular respiration, generating energy for the cell. The cell needs to be large enough to house not only many mitochondria (to meet its energy demands) but also other vital organelles like the nucleus, chloroplasts (in plant cells), vacuoles, and endoplasmic reticulum. This significant size difference allows for the compartmentalization and efficient functioning of all these components within the cell.